Software Requirements and Implementation Plan: KhadSetu v1

Team [NAME] | One district in v1 | AI proposes. Rules dispose. Humans decide.

- Scope: stock-backed tokens matched to sowing dates and nearest feasible centre, inclusion fallbacks, simulator-tested rules, four bounded AI assists.

- Out of scope in v1: real AgriStack/iFMS/SMS, payments, partial fulfilment, statewide scale.

1. Users and roles

User	Who	Needs
Farmer	Landowner / tenant / assisted	Token before sowing; Hindi; no dead ends
Centre operator	Retailer staff	Redeem, lots, audits, assisted registration
Village verifier	Patwari / sarpanch	Tenant queue, daily cap 40
District officer	Agri officer	Stock view, flags, rule testing

2. Inputs

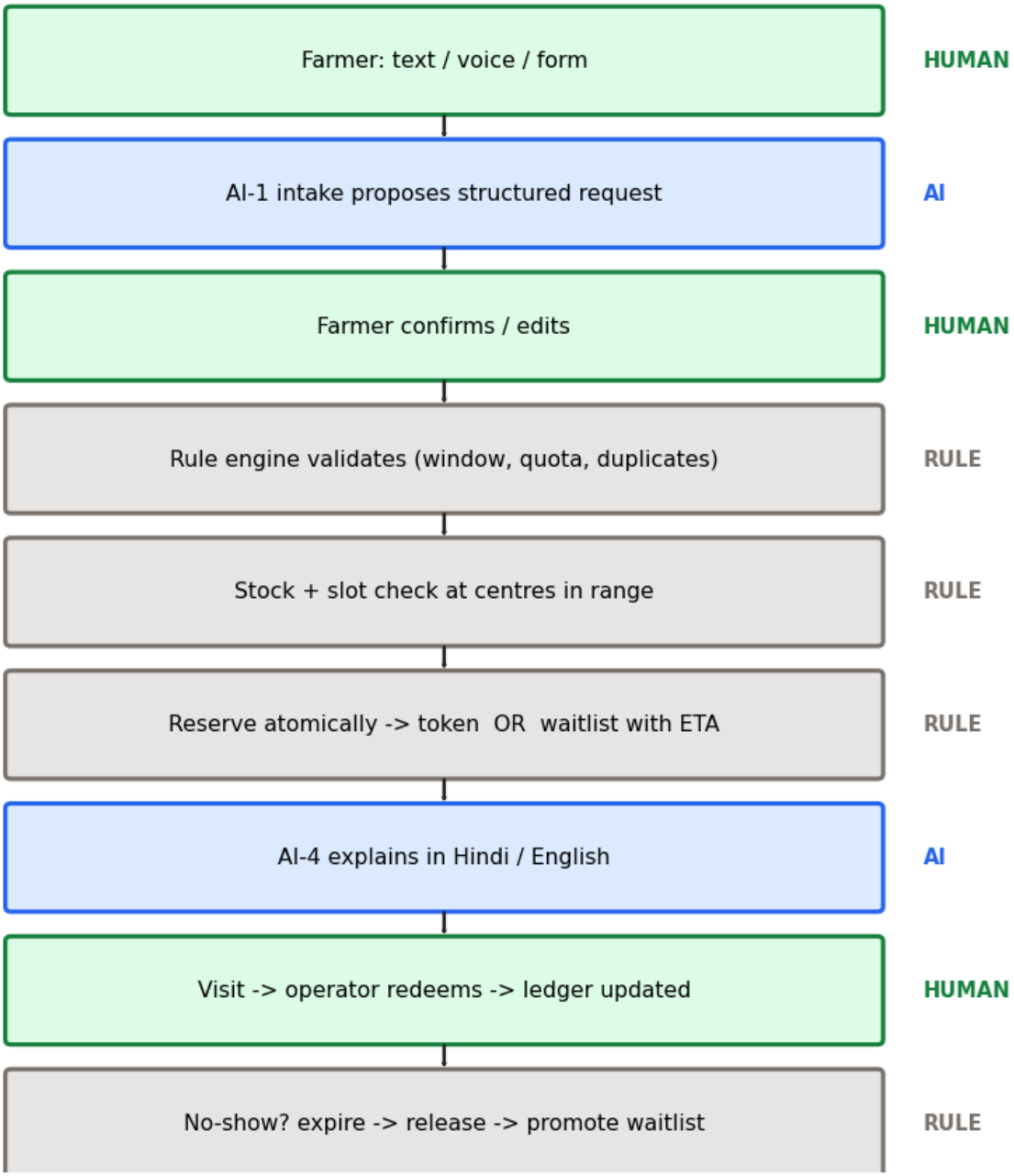
User	Input	Validation
Farmer	Phone, name, village, acres, product, crop, qty, sowing date (text/voice/form)	Crop window; entitlement cap; unique phone
Tenant	Declared acres, village, optional landowner	Provisional until verified
Operator	Token code, lots, counted stock	Code valid + in date; numeric counts
Verifier	Approve / reject + note	Daily approval limit
Officer	Rule values, sim parameters	Allowed ranges; allowlisted products

3. Outputs

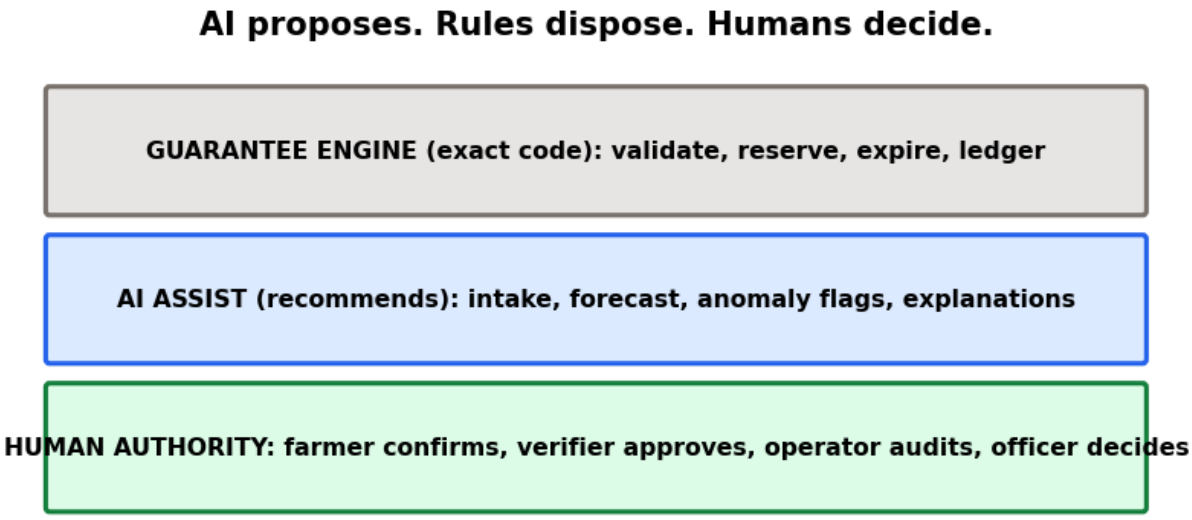
Token (code, QR, centre, date, slot, expiry); waitlist status with ETA; plain-language explanation (Hindi/English); operator dashboard; officer dashboard (health, demand vs supply, at-risk, flags); simulator comparison report; decision provenance timeline.

4. Workflow

Token workflow: blue = AI, grey = rules, green = humans



Swimlanes: Farmer | AI (blue) | Rule engine (grey) | Humans (green). Steps: register (OTP or fallback; tenants provisional) > state need (AI-1 drafts) > farmer confirms > engine validates > stock+slot check > reserve+token or honest waitlist > AI-4 explains > redeem and ledger update > no-show expiry and promotion > audits with reassignment > officer monitoring and rule tests.



5. Functional requirements

ID	Requirement	Pri	Acceptance check
FR-01	Token only after atomic reservation	Must	100 concurrent on 50 bags: at most 50 tokens
FR-02	Pickup before sowing unless urgent	Must	No token on/after sowing unless flagged
FR-03	Nearest feasible centre within detour	Must	Next-nearest chosen when nearest dry
FR-04	Waitlist with ETA when no stock	Must	ETA + at-risk flag, no token
FR-05	Expire no-shows, release + promote	Must	Stock returns, top waiter promotes
FR-06	One free reschedule	Should	Second reschedule refused
FR-07	Tenant provisional path (60% then 100%)	Must	Quota steps up on approval
FR-08	OTP fallbacks	Must	100% OTP failure still registers
FR-09	Redeem by code/QR	Must	Stock + ledger update
FR-10	Audit with discrepancy log	Should	Shortfall logged, tokens reassigned
FR-11	Simulator FIFO vs KhadSetu, seeds	Must	Mean, spread, where-we-lose
FR-12	Constraint Console re-runs sim	Should	Both systems shown < 20 s
FR-13	Decision provenance per token	Must	AI / rule / human steps visible
FR-14	Officer dashboard + flags	Should	Updates after clock advance

6. AI responsibilities

ID	AI does	Input	Output	Guardrail	Override
AI-1	Text/voice to structured request	Farmer message	Request + confidence	Schema + re-validation; asks 1 question; cannot submit	Farmer confirms/edits; form fallback

ID	AI does	Input	Output	Guardrail	Override
AI-2	Demand forecast per centre/week	Acres, calendar, rain signal	Forecast + interval	MAE vs naive shown; planning only	Officer moves stock
AI-3	Anomaly flags	Approvals, gaps, no-shows	Ranked flags + reasons	Flags only; precision/recall measured	Officer reviews/dismisses
AI-4	Explains + officer brief	Decision record	Plain text	Paraphrase only; fidelity-checked; template fallback	Officer edits brief

AI is NEVER allowed to: reserve/release stock, approve eligibility or tenants, penalise anyone, or submit a request.

7. Human responsibilities

ID	Who	Decision	Why a human
H-01	Farmer	Confirms every request	Their commitment and data
H-02	Verifier	Tenant approve/reject	Local knowledge, accountability
H-03	Operator	Counts stock, redeems	Physical reality decides
H-04	Officer	Rules + redistribution	Policy and fairness
H-05	Officer	Reviews AI flags	Suspicion is not proof
H-06	Team	Accepts AI outputs	Judgment layer

8. Constraints

Low connectivity; Hindi-first and no-smartphone users; stock accuracy depends on operators; synthetic data until pilot; 36-hour build; farmer-data privacy; OTP/Farmer ID/SMS/stock feeds mocked in v1.

9. Failure handling

Failure	Detection	Fallback	Owner
Wrong stock figures	Audit gap	Buffer; reassign tokens	Operator
OTP down	Verify errors	Voice / assisted	Operator
No-shows	Past grace date	Release; promote	System
Centre closed	Officer marks inactive	Re-run allocation	Officer
AI wrong/down	Low confidence/timeout	Form + templates	Farmer/system
Verifier misuse	Anomaly flag	Review; lower quota	Officer
Demand spike	Forecast alert	Waitlist + redistribution	Officer

10. Acceptance criteria (Given / When / Then)

1. 50 bags + 100 simultaneous 1-bag requests > at most 50 tokens issued.
2. Nearest centre dry > token at next-nearest within detour.
3. No stock in range > waitlist with ETA, no token.
4. Token past date+grace + clock advances > expires, releases, top waiter promotes.
5. Tenant without Farmer ID registers > provisional token at 60% quota.
6. OTP fails twice + fallback chosen > registration completes, flagged.
7. Hindi message + low AI confidence > question or form shown, nothing submitted.
8. AI-4 explanation vs decision record > zero facts outside the record.
9. Supply cut 30% in console + run > both systems shown within 20 s.
10. Any token opened > provenance shows AI / rule / human steps.
11. Audit shortfall submitted > discrepancy logged, tokens reassigned.

11. Version-1 exclusions

Real Aadhaar / AgriStack / iFMS / SMS / payments; partial fulfilment; multi-product baskets; statewide scale; production security; real-time GPS.

12. Adaptability playbook (judge changes)

Judge says	Change	Check
Cut stock 30%	Supply parameter	Waitlist growth; wasted trips
OTP is down	OTP rate to 100%	Fallback registration works
Add a product	Config + norms	Request succeeds
Lower per-farmer cap	Cap scale	Fairness before/after
Close a centre	Mark inactive	Re-assignment + travel delta
Prioritise small farmers	Waitlist weights	In-window by holding size